1. More Conditional Probability

Q: A test for detecting cancer has the following accuracy

1. if a person has cancer, there is a 10% chacne that the test will say that the person does not have it. This is called a "false negative."
2. If a person does not have cancer, there is a 5% chance that the test will say that the person does have it. This is called a "false positive."

For patents that have no family history of cancer, the incidence of cancer is 1%. Person X does not have any family history of cancer, but is detected to have cancer. What is the probability that the Person X dose have cancer?

$S = \{(\text{Healthy}, \text{Positive}), (\text{Healthy}, \text{Negative}), (\text{Sick}, \text{Positive}), (\text{Sick}, \text{Negative})\}$

A is the event that Person X has cancer. B is the event that the test is positive.

$$P[A|B] = \frac{P[A \cap B]}{P[B]} = \frac{P[\{(S,P)\}]}{P[\{(S,P),(H,P)\}]} = \frac{0.0090}{0.0090 + 0.0495} \approx 15.4\%$$

Bayes Rule

For events E and F if $P[E] \neq 0$ and $P[F] \neq 0$, then $P[E|F] = \frac{P[E|F]P[F]}{P[E]}$

Proof: Using the formula for conditional probability,

$$\begin{aligned}
\Pr[E|F] &= \frac{\Pr[E \cap F]}{\Pr[F]} \quad ; \quad \Pr[F|E] = \frac{\Pr[F \cap E]}{\Pr[E]} \\
\implies \Pr[E \cap F] &= \Pr[E|F]\Pr[F] \quad ; \quad \Pr[F \cap E] = \Pr[F|E]\Pr[E] \\
\implies \Pr[E|F]\Pr[F] &= \Pr[F|E]\Pr[E] \\
\implies \Pr[F|E] &= \frac{\Pr[E|F]\Pr[F]}{\Pr[E]}
\end{aligned}$$

2. The Law of Total Probability and Bayes rule

For events E and F, $P[E] = P[E|F]P[F] + P[E|F^c]P[F^c]$

Proof:

$$E = (E \cap F) \cup (E \cap F^c)$$
$$\implies P[E] = P[(E \cap F) \cup (E \cap F^c)] = P[E \cap F] + P[E \cap F^c]$$
$$P[E] = P[E|F]P[F] + P[E|F^c]P[F^c]$$

1. Apply the union rule for mutually exclusive events.
2. By the definition of conditional probability.

If we **combine** bayes rule and the Law of Total Probability, we get

$P[F|E] = \frac{P[F \cap E]}{P[E]} = \frac{P[E|F]P[F]}{P[E]}$ (By definition of conditional probability)

$P[F|E] = \frac{P[E|F]P[F]}{P[E|F]P[F] + P[E|F^c]P[F^c]}$ (By the law of total probability)

Q: Provce that for events E and F, $P[E^c|F] = 1 - P[E|F]$.

Proof: Since $E \cup E^c = S$, for any event F s.t. $P[F] \neq 0$,

$$(E \cup E^c) \cap F = S \cap F = F$$
$$(E \cup E^c) \cap F = (E \cap F) \cup (E^c \cap F)$$
$$\implies P[(E \cap F) \cup (E^c \cap F)] = P[F]$$

Since $E \cap F$ and $E^c \cap F$ are mutually exclusive,

$$P[E \cap F] + P[E^c \cap F] = P[F] \implies \frac{P[E^c \cap F]}{P[F]} = 1 - \frac{P[E \cap F]}{P[F]}$$
$$\implies P[E^c|F] = 1 - P[E|F]$$

Q: